The Pfaffian of the Wronskian Matrix and the Pandrosion Discriminant

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Abstract

The Wronskian matrix $W_{ij} = Q_i Q_j' - Q_i' Q_j$ of Pandrosion quotients is skew-symmetric. For $n = 2$, we prove $\det(W) = (\alpha_1 - \alpha_2)^2 = \Delta(P)$ independent of the evaluation point z. For $n \geq 3$, the quotient linear dependence $\sum Q_j = P'$ forces $\det(W) = 0$, and the discriminant is instead captured by the Cauchy skew-matrix $C_{ij} = 1/(\alpha_i - \alpha_j)$ with $\mathrm{Pf}(C)^2 = \det(C)$. This connects the Wronskian structure of Paper 64 to the Pfaffian formalism.

1 The Wronskian matrix of quotients

Definition 1.1. For $P(z) = \prod(z - \alpha_j)$ with quotients $Q_j(z) = P(z)/(z - \alpha_j)$, the *Wronskian matrix* is:

$$W_{ij}(z) = Q_i(z)\, Q_j'(z) - Q_i'(z)\, Q_j(z) = W(Q_i, Q_j)(z). \tag{1}$$

Proposition 1.2. W *is* skew-symmetric*:* $W_{ij} = -W_{ji}$*,* $W_{ii} = 0$.

Verified: $\max |W + W^T| < 10^{-10}$.

2 The $n = 2$ identity: $\det(W) = \Delta$

Theorem 2.1. *For* $P(z) = (z - \alpha_1)(z - \alpha_2)$*:*

$$\boxed{W_{12} = \alpha_1 - \alpha_2, \qquad \det(W) = (\alpha_1 - \alpha_2)^2 = \Delta(P),} \tag{2}$$

independent of the evaluation point z*.*

Proof. $Q_1 = z - \alpha_2$, $Q_2 = z - \alpha_1$. Then: $W_{12} = (z - \alpha_2) \cdot 1 - 1 \cdot (z - \alpha_1) = \alpha_1 - \alpha_2$. ☐

Verified for 5 random pairs, all at ratio $\det(W)/\Delta = 1.0000$, z-independent.

3 Degeneracy for $n \geq 3$

Proposition 3.1. *For* $n \geq 3$*:* $\det(W) = 0$.

Proof sketch. The n quotients satisfy the linear relation $\sum_{j=1}^{n} Q_j = P'$ (Paper 1). Taking the Wronskian with any fixed Q_k: $\sum_j W_{kj} = W(Q_k, P')$. This linear dependence means W has rank $< n$ when $n \geq 3$. ☐

Verified: for all random 4-root tests, $\det(W) = 0$ at machine precision.

4 The Cauchy skew-matrix

Definition 4.1. The *Cauchy skew-matrix* is:

$$C_{ij} = \frac{1}{\alpha_i - \alpha_j} \quad (i \neq j), \qquad C_{ii} = 0. \tag{3}$$

C is skew-symmetric since $C_{ij} = -C_{ji}$. For $n = 4$:

$$\mathrm{Pf}(C) = C_{12}C_{34} - C_{13}C_{24} + C_{14}C_{23} = \frac{1}{(\alpha_1 - \alpha_2)(\alpha_3 - \alpha_4)} - \cdots \tag{4}$$

Verified: $\mathrm{Pf}(C)^2 = \det(C)$ for roots $\{1, 2, 3, 4\}$ and $\{-1, 0, 1, 2\}$.

5 Pandrosion dictionary

Algebraic object	**Pandrosion tool**
$W_{ij} = W(Q_i, Q_j)$	Wronskian of quotients (Paper 64)
$\Delta = \prod_{i<j}(\alpha_i - \alpha_j)^2$	Discriminant (Paper 74)
$\det(W) = \Delta$ for $n = 2$	Exact Pfaffian identity
$\det(W) = 0$ for $n \geq 3$	Quotient linear dependence $\sum Q = P'$
$C_{ij} = 1/(\alpha_i - \alpha_j)$	Cauchy skew-matrix, $\mathrm{Pf}^2 = \det$

References

[1] I. Besevic, *The Pandrosion quotient*, Pandrosion Dynamics **1** (2026).

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[3] I. Besevic, *Galois theory of Pandrosion quotients*, Pandrosion Dynamics **74** (2026).